1 SLIDE 01

KARNATAKA STATE POLICE

PROJECT NETRA

AI INTELLIGENCE PLATFORM FOR KARNATAKA STATE POLICE

Empowering Law Enforcement with Trusted Agentic AI, Multilingual Intelligence, and Rapid Crime Data Analysis.

- AI-POWERED INTELLIGENCE
- KANNADA + ENGLISH
- VERIFIED INFORMATION
- RAPID SEARCH & ANALYSIS
- SECURE & RELIABLE







KSP DATATHON 2026

INNOVATE • INTELLIGENT • IMPACTFUL

TEAM CREDENTIALS

CYBER EYE INTELLIGENCE

KARNATAKA STATE POLICE
KSP



PROJECT LEAD

Shridhar Madival

Driving the vision and end-to-end development of Project Netra.



TEAM MEMBER

Siddu S

Backend development, database integration, and system architecture.



TEAM MEMBER

Neha Saini

Frontend development, UI/UX design, and user experience.

PROBLEM STATEMENT

Challenges Faced by Law Enforcement

KARNATAKA STATE POLICE

DIFFICULT TO SEARCH CRIME RECORDS

Crime records are difficult to search quickly.

BILINGUAL DATA CHALLENGE

Officers work with both Kannada and English data.

LEGACY SEARCH SYSTEMS

Legacy search systems are slow and keyword-dependent.

UNRELIABLE AI SUMMARIES

Traditional AI may generate unreliable or hallucinated summaries.

NEED VERIFIED INFORMATION

Investigators need verified, structured information for decision-making.

IMPACT

- INCREASED INVESTIGATION TIME
- HIGHER MANUAL EFFORT
- DELAYED ACCESS TO HISTORICAL CASE INFORMATION

Project Netra addresses these challenges by combining multilingual intelligence, verified database search, and AI-powered insights into a single unified platform.

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ROLE-BASED SECURE INTELLIGENCE ACCESS

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Separate Database & Dashboards for DGP, SP & PSI

Project Netra uses **hierarchical, role-based** secure databases ensuring every police rank only accesses information relevant to their **operational authority**.

DGP DASHBOARD

State-Level Intelligence

DGP DATABASE
State-Wide Data

- State-wide crime overview
- Crime heatmaps & trends
- Cross-district analytics
- Strategic monitoring & decisions

SP DASHBOARD

District-Level Management

SP DATABASE
District Data

- District-level crime management
- Officer performance tracking
- District investigations
- Resource allocation & control

PSI DASHBOARD

Station-Level Operations

PSI DATABASE
Station Data

- Station-specific FIR search
- Case investigation & updates
- Suspect lookup & records
- Voice-based AI assistant

SECURITY & ACCESS CONTROL

- Hierarchical Access Control
- Zero Unauthorized Data Exposure
- Encrypted Role-Based Databases
- Audit Logs & Activity Monitoring

KEY BENEFITS

- Right Data to Right Officer
- Faster Decisions at Every Level
- Enhanced Security & Data Integrity
- Efficient Policing & Better Outcomes

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OUR SOLUTION
PROJECT NETRA

KARNATAKA STATE POLICE



An AI-powered intelligence platform that enables Karnataka Police officers to query crime records naturally in Kannada or English, retrieve verified case information from the Catalyst database, and receive concise, structured AI-generated responses while minimizing hallucinations.

CORE CAPABILITIES

- Bilingual natural-language search
- Intelligent keyword extraction
- Catalyst + ZCQL data retrieval
- QuickML-powered summarization
- Voice input support
- Tactical dashboard with session history

Officer → Voice / Text Query → NLP Processing → ZCQL Search → QuickML Summary → AI Summary → Voice Support → Tactical Dashboard

Project Netra brings together multilingual intelligence, verified data, and agentic AI to deliver **fast**, **reliable**, and **actionable** insights for Karnataka Police.

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CORE FEATURES

Intelligent Crime Intelligence Platform

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01

BILINGUAL SEARCH

Search crime records using Kannada or English.

02

VOICE-TO-TEXT

Supports kn-IN and en-IN speech input.

03

AGENTIC KEYWORD EXTRACTOR

Custom NLP engine removes filler words and understands regional language variations.

04

PARALLEL DATABASE SEARCH

Fast retrieval using Catalyst ZCQL with a JavaScript fallback scanner.



05

AI CASE SUMMARY

QuickML generates concise, structured summaries from retrieved records.

06

ACCUSED INFORMATION EXTRACTION

Extracts key accused details from verified case data.

07

SESSION HISTORY

Stores previous searches locally for quick reference and reuse.

08

GOVERNMENT-READY DASHBOARD

Clean, secure dashboard designed for law enforcement workflows.



Project Netra empowers officers with intelligent, verified, and multilingual AI for faster and smarter crime investigations.

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PROJECT NETRA

SYSTEM ARCHITECTURE

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POLICE OFFICER
User interacts via dashboard using voice or text.

VOICE / TEXT INPUT
Voice (kn-IN / en-IN) or manual text query.

CUSTOM NLP ENGINE
Language detection and intelligent keyword extraction.

ZCQL SEARCH
Primary search using Catalyst ZCQL.

JS BACKUP SCANNER
JavaScript fallback scanner for validation and coverage.

CATALYST SERVER
Node.js + Express (Serverless Functions)

MERGED & VERIFIED RESULTS
Combine and verify results from both search methods for accuracy.

ZOHO QUICKML
Summarization and information extraction from case data.

TACTICAL DASHBOARD
Display AI summary, key details and store search locally.

TECHNOLOGY STACK
- Node.js
- Express.js
- Zoho Catalyst
- ZCQL
- Zoho QuickML
- JavaScript
- Web Speech API

KEY FEATURES
- Bilingual Search (Kannada + English)
- Voice-to-Text Support (kn-IN / en-IN)
- Agentic Keyword Extractor
- Parallel Database Search
- AI Case Summary
- Accused Information Extraction
- Session History
- Government-Ready Dashboard



Project Netra connects **multilingual intelligence**, verified data, and AI to empower Karnataka Police with **faster and smarter** crime investigations.

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HOW PROJECT NETRA WORKS

From Query to Intelligence in Seconds

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1 USER QUERY

Officer submits a query in Kannada or English using text or voice.

ಉದಾಹರಣೆ: "ಚಿಕಮಗಳೂರು ನಲ್ಲಿ ಕಳ್ಳತನ ಪ್ರಕರಣಗಳು"

Example: "Theft cases in Chikkamagaluru"

2 INPUT PROCESSING

Voice is converted to text and the query is cleaned and processed.

3 INTELLIGENT KEYWORD EXTRACTION

NLP engine extracts meaningful keywords and understands intent & context.

4 DATABASE SEARCH

System searches Catalyst Data Store using ZCQL with optimized queries.

5 DATA VALIDATION & RETRIEVAL

Relevant records are validated and retrieved from the database securely.

6 AI SUMMARIZATION (QUICKML)

Zoho QuickML generates concise, structured summaries from case records.

7 RESPONSE TO OFFICER

AI summary and key case details are shown on the dashboard instantly.

Officer → Query → Processing → NLP & Keywords → Database Search → Validation → AI Summary → Dashboard



Project Netra intelligently connects natural language queries with verified database records and AI-generated insights to deliver **fast and reliable crime intelligence.**

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KEY BENEFITS OF PROJECT NETRA

Smarter Intelligence. Stronger Enforcement.

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01 FASTER SEARCH & RETRIEVAL

Find relevant case records in seconds using intelligent keywords and ZCQL search.

02 MULTILINGUAL INTELLIGENCE

Supports Kannada and English for natural, bilingual interaction.

03 AI-GENERATED SUMMARIES

QuickML provides concise, accurate case summaries with key insights.

04 ACCURATE & VERIFIED DATA

Information is retrieved from the verified Catalyst database ensuring accuracy.

05 REDUCED MANUAL WORKLOAD

Automates search, filtering and summarization—saving officers valuable time.

06 BETTER DECISION MAKING

Structured insights and historical data empower officers to act confidently.

07 SECURE & RELIABLE

Built with secure authentication and role-based access for safe data handling.

08 SCALABLE & FUTURE READY

Modular architecture allows easy integration of new features and data sources.



OVERALL IMPACT

Faster investigations | Improved accuracy | Empowered officers | Safer communities

Project Netra transforms the way Karnataka Police accesses and understands crime data – **faster, smarter, and more effectively.**



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FUTURE ROADMAP OF PROJECT NETRA

Building a Smarter, Safer & AI-Driven Karnataka

KARNATAKA STATE POLICE — KSP —

PHASE 1

FOUNDATION

Core platform development, Catalyst integration, ZCQL search, and bilingual query support.

COMPLETED

PHASE 2

AI ENHANCEMENT

Improve NLP accuracy, expand keyword engine, and integrate QuickML for advanced summarization.

IN PROGRESS

PHASE 3

ADVANCED ANALYTICS

Add case trend analysis, crime pattern detection, and AI-powered recommendations for better decisions.

NEXT

PHASE 4

SYSTEM INTEGRATION

Integrate with other police systems (CCTNS, e-FIR, ICJS) for unified data access and real-time information flow.

UPCOMING

PHASE 5

AI AUTONOMY

Towards autonomous AI agents, predictive policing, and voice assistants for proactive crime prevention.

FUTURE

PROJECT NETRA IN NUMBERS

- **150,000+** FIR Records (Integrated & Growing)
- **2** Languages Supported (Kannada & English)
- **< 3 Seconds** Average Search & Response Time
- **100%** Verified Data from Catalyst Database
- **Empowering** Police Officers Across Karnataka

OUR VISION

To make Project Netra the most advanced AI-powered crime intelligence platform in India, enabling Karnataka Police to prevent crimes, solve cases faster, and serve citizens better.

TECHNOLOGY WE WILL CONTINUE TO LEVERAGE

Zoho Catalyst | ZCQL | QuickML | Node.js | Express.js | JavaScript | Web Speech API



Project Netra is not just a platform — it is the future of intelligent policing in Karnataka.

Smart Today. Safer Tomorrow.

THANK YOU!

Project Netra – Intelligent. Verified. Multilingual.

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PROJECT NETRA

Cyber Eye • Intelligent Crime Intelligence Platform

Q&A

We're here to build a
Smarter, Safer Karnataka.

CONNECT WITH US

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helpdesk@ksp.gov.in

Phone
080-2294 4444

Address
Karnataka State Police HQ,
Bengaluru, Karnataka – 560001

Together, we can fight crime smarter!

INTELLIGENT
AI-powered insights for faster decisions.

VERIFIED
Accurate data from Catalyst database.

MULTILINGUAL
Kannada & English support.

SECURE
Built with security and integrity.

EMPOWERING
Empowering officers and communities.

> "Intelligence leads to prevention. Technology leads to transformation.
>
> **Let's build a safer tomorrow.**



THANK YOU

For your time and support.
Together, let's make Karnataka a safer place for everyone.